

From LLMs to Agentic AI: How Recent Advances Can Enhance Our Daily Work

Subtitle:

From prompting and AI-assisted coding to tools, agents, and multi-agent workflows in dentistry, research, and healthcare

Course overview

This four-part seminar introduces students to the rapid evolution of generative AI—from large language models (LLMs) and conversational AI to agentic AI systems that can use tools, work through multi-step tasks, and coordinate specialized AI agents.

Rather than focusing only on how to write better prompts, the course explores a more fundamental shift: from asking AI to generate an answer to designing AI systems that can help accomplish a task.

Students will first develop an intuitive understanding of how LLMs work, including tokens, transformers, training, context, and tool use. They will then explore how LLMs become agentic systems when combined with tools, memory/state, planning, feedback, and an execution environment.

The course will emphasize hands-on experimentation with realistic tasks relevant to dentistry and academic work, including literature research, information synthesis, data analysis, document creation, and AI-assisted programming and prototyping. Students will also experiment with multi-agent systems, where specialized agents collaborate under the coordination of a manager agent.

The goal is to give students the conceptual understanding and practical skills needed to recognize where agentic AI can augment their work, design useful AI workflows, and critically evaluate their outputs and limitations.

By the end of the course, students will be able to:

1. Explain the basic principles behind LLMs

Understand tokens, embeddings, transformers, training, inference, context windows, and why LLMs can generate surprisingly capable responses without functioning like conventional databases or search engines.

2. Understand the transition from LLMs to agentic AI

Explain how an LLM can be embedded in a system with tools, memory/state, planning, feedback, and an execution environment to perform multi-step tasks.

3. **Distinguish between an LLM, an AI assistant, an agent, and a multi-agent system**

Understand what changes when AI moves from generating text to selecting tools, executing actions, observing results, and coordinating other specialized agents.

4. **Use AI for practical research and knowledge-work tasks**

Experiment with AI workflows for literature discovery, paper analysis, evidence synthesis, data processing, document generation, and other academic tasks relevant to dentistry.

5. **Use AI-assisted programming and “vibe coding” to prototype solutions**

Learn how modern AI coding tools can help students build small scripts, data-analysis workflows, and functional prototypes without requiring them to become professional programmers.

6. **Design and experiment with multi-agent workflows**

Understand how specialized agents can be explicitly configured with different roles, instructions, tools, and responsibilities, and how a manager agent can coordinate them to solve a larger task.

7. **Critically evaluate AI-generated results and agentic workflows**

Recognize hallucinations, unsupported claims, automation errors, inappropriate tool use, and the limitations of AI systems in research and healthcare.

8. **Identify appropriate applications of agentic AI in dentistry and healthcare**

Evaluate where AI can potentially augment clinical, educational, administrative, and research work—and where human expertise, verification, and responsibility must remain central.